AI02

Research Proposal Presentation

Balancing Accuracy and Interpretability in Deep Explainable Recommendation Systems: A Literature Review

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Agenda

01 Introduction

02 Problem Statement

03 Method Overview

04 Key Insights & Research Gaps

05 Conclusion



Introduction

- Recommender Systems (RS) are widely used in e-commerce, social media, streaming.
- Deep Learning improves recommendation accuracy significantly.
- However → models become black-box (hard to explain).
- Lack of transparency reduces user trust and system reliability.

→ ***Problem: Need systems that are both accurate AND explainable***

Motivations

AI's function has changed from being an expensive item to a basic requirement for meaningful information discovery as digital data develops at an exponential rate.



Data Volume

With millions of products and users, manual discovery is impossible; users now rely entirely on AI filtering.



Deep Learning Advantage

Modern DL models identify complex user-item connections that traditional methods miss.



Knowledge Integration

Using Graph Neural Networks (GNNs) and Knowledge Graphs (KGs) to incorporate real-world data.



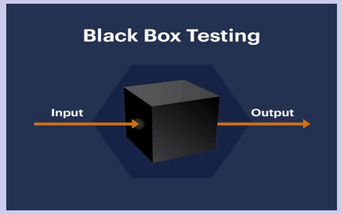
User Benefits

Enhances experiences for new users and improves reliability in sensitive fields like E-commerce and Healthcare.

Research Gaps

Introduction

01. The "Black-Box" Problem



02. The Accuracy-Interpretability Trade-off



03. The Missing Link



Proposed Method

Main Goal: To create a system that is both highly accurate (gives the right advice) and easy to understand (explains why).



Smart Filtering

Like a personal assistant, the system picks out only the most relevant facts from a massive database for each user.



Logic Tracing

An AI "guide" finds a clear path of connections—like a map—to show exactly how a user is connected to a recommended item.



Balanced Optimization

The system is trained to reach two goals at once: staying precise while making its explanations clear and helpful.



Structural Alignment

Ensure the explanation given to the user is the actual reason the AI made that choice, not just a guess.

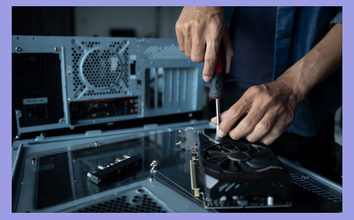
Key Contributions

Introduction

01. Unified Classification



02. The Accuracy Gap



03. Real-World Application



04. Future Roadmap



Problem Statement

A key challenge in modern recommender systems is balancing predictive accuracy with model interpretability.



Accuracy >< Interpretability Trade-off

Deep models → High accuracy, Low interpretability

Simple models → High interpretability, Low accuracy



No unified solution exists

No single method can consistently achieve both high accuracy and strong interpretability.



No standard evaluation for interpretability

There is no common metric to objectively measure interpretability across models.

Related Work

Recommender Systems

Collaborative Filtering (CF)

Content-based Filtering

Hybrid

Deep Learning

MLP, CNN, RNN, Attention

GNN (Graph Neural Network)



Method Overview



- **Core Ideal**

→ KG Encoder + Path Decoder

- **General Structure**

→ Provide reasoning paths (high interpretability)

- **Multi-objective optimization**

→ Optimize accuracy + interpretability simultaneously

- **Causal / Counterfactual methods**

→ Explain “why” and “what-if”

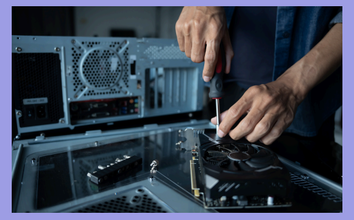
Key Insights & Research Gaps

Research Gaps

01. No unified framework balancing both accuracy & interpretability



02. Lack of standard metrics for explainability & user trust



03. Limited real-world / user-based evaluation



04. Scalability issues in complex models



Recommendation Methods

How does AI recommend products, movies, or videos?

And how does it explain those decisions?

Accuracy (độ chính xác)

Interpretability (khả năng giải thích)

Attention-based Methods

Attention → Focus on important signals

Knowledge Graph-based Methods

Attention → Focus on important signals

Multi-objective Optimization

Multi-objective → Balance multiple goals

Post-hoc Explanation

Post-hoc → Explain after prediction

Attention-based Methods

Uses the attention mechanism to identify the most relevant information.
Assign weights to user behavior or product characteristics.
Help the system focus on strong signals when making recommendations.

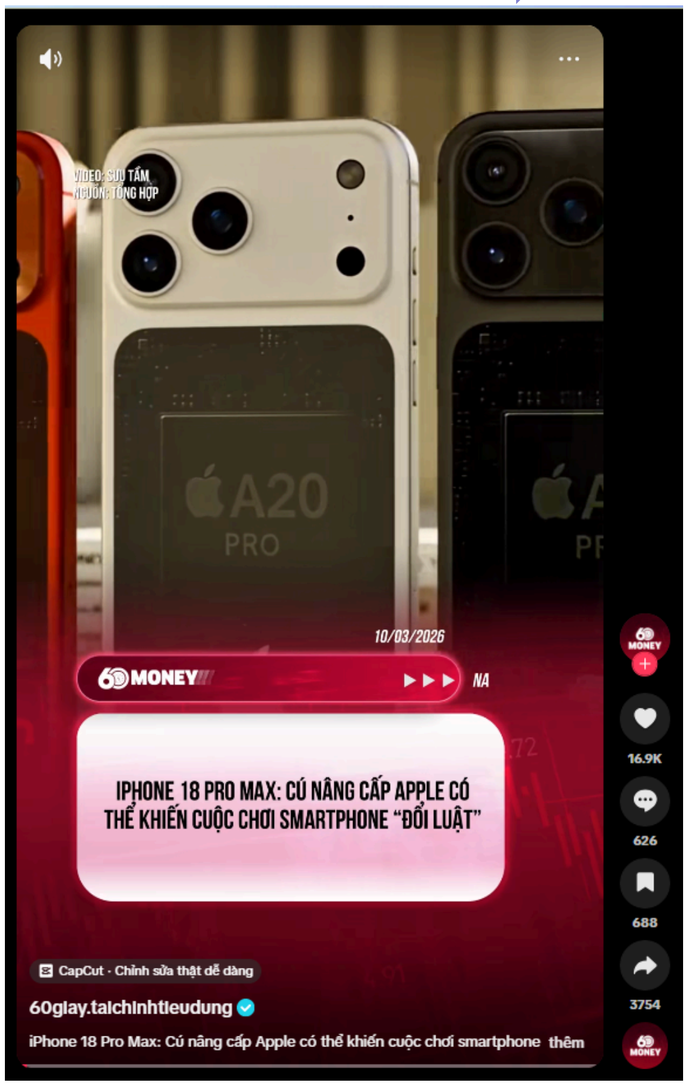
Advantages

- High accuracy
- Fast inference speed
- Integrated with Deep Learning

Disadvantages

- Only focuses on the importance of the data
- May misinterpret true user intentions
- Limited explanation depth

"Attention sees behavior, but doesn't necessarily understand the person."



Knowledge Graph-based Methods

Knowledge Graph-based Methods use structured relationships among users, items, and attributes to generate recommendations with interpretable reasoning paths.

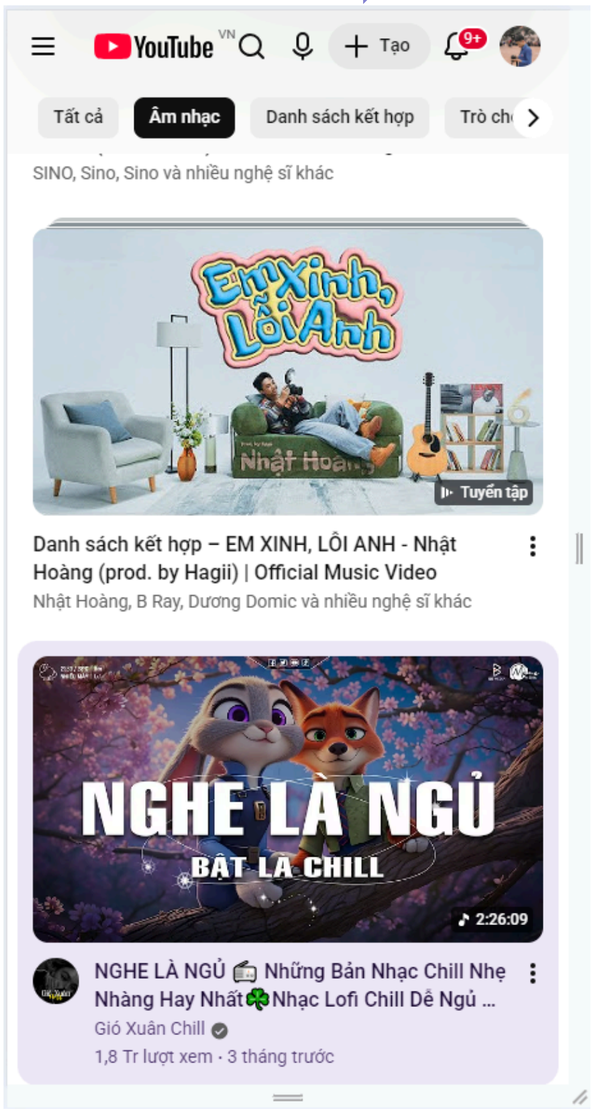
Advantages

- Clear and human-readable explanations
- Improves user trust
- Helps cold-start recommendation

Disadvantages

- Difficult to build and maintain graph data
- High computational cost
- Hard to update in real-time

"Knowledge Graphs were created to help AI understand the relationships between data."



Multi-objective Optimization

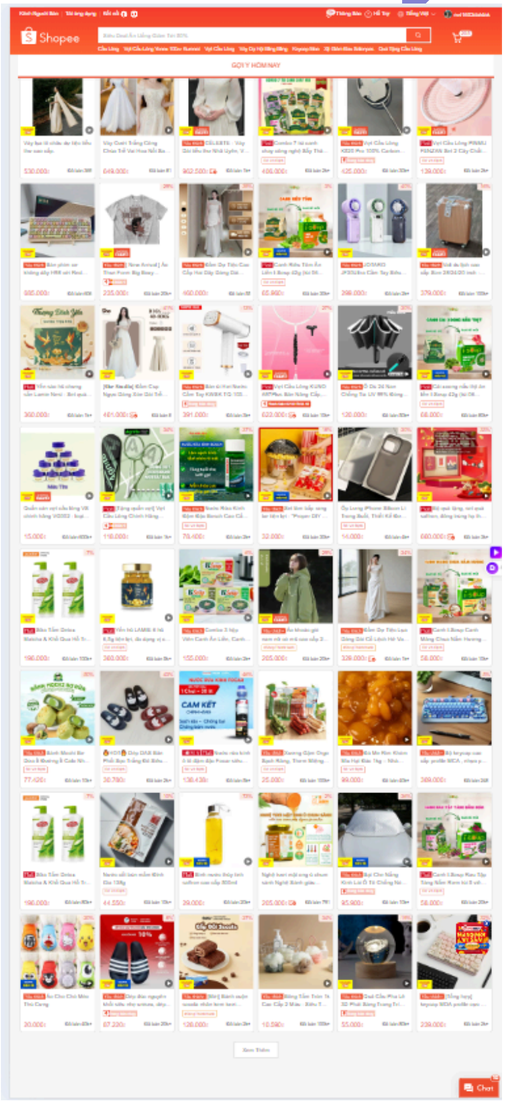
Multi-objective Optimization is a method that helps AI optimize multiple objectives simultaneously, such as providing accurate recommendations while also being able to explain them.

Advantages

- AI can accurately explain clearly
- Better suited for practical applications
- Supports multiple business goals

Disadvantages

- Hard to satisfy all goals at the same time
- Requires more time and resources to train
- Needs careful adjustment of priorities



"Multi-objective optimization is the way AI tries to excel at many things at the same time, instead of just one thing."

Conclusion

Deep RS → high performance but lack transparency

- Capture complex user–item patterns with deep neural networks.

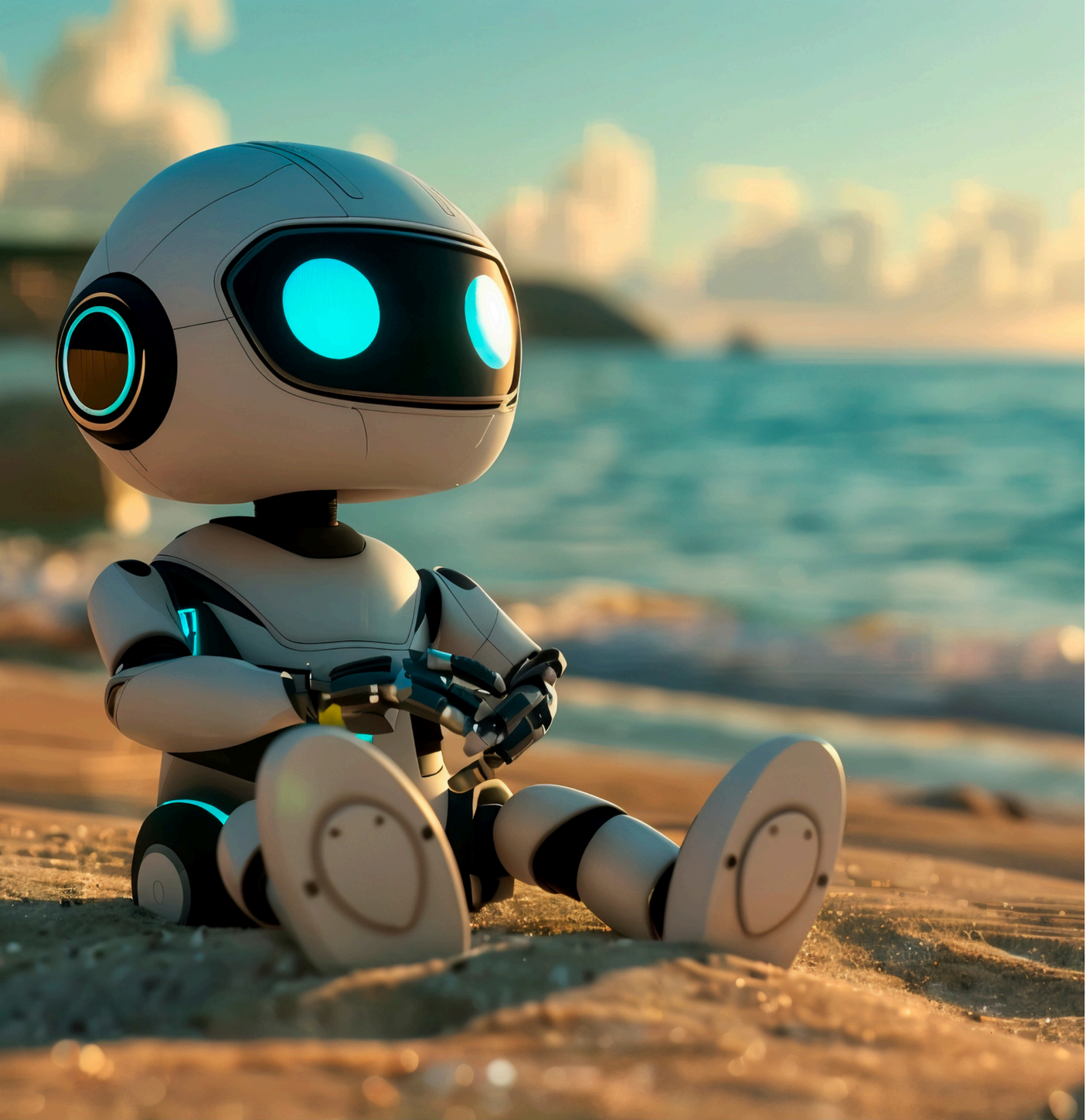
Explainable methods improve trust but may reduce accuracy

- Provide human-understandable explanations (e.g., paths, attention).

Trade-off remains an open research challenge

- Improving one objective often degrades the other.





Future Direction

Integrate LLMs for natural explanations

- Generate human-like, textual explanations for recommendations.

Develop human-centered evaluation metrics

- Measure user trust, satisfaction, and explanation usefulness.

Design scalable hybrid models (DL + symbolic reasoning)

- Combine deep learning with structured reasoning (e.g., KG paths).

Thank You